AMRITSAR GROUP OF COLLEGES

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Major Project Report

On

"Smart Pukhraj 26.0"

Submitted in the partial fulfilment of the requirement for the award of degree of

Bachelor of Technology

In

COMPUTER SCIENCE & ENGINEERING

Batch (2022-2026)



Submitted To:

Department of CSE

Submitted By:

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I, **Kanwaldeep Singh (Roll No. 2233626)**, student of **Final Year B.Tech (Computer Science & Engineering)** at **Amritsar Group of Colleges**, hereby declare that the project work entitled **"Smart Pukhraj 26.0"**, submitted to **Amritsar Group of Colleges** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology**, is the result of our own work carried out under the valuable guidance of **Er.Tejinder Sharma (Associate Professor)**.

We further declare that this project has not been submitted to any other institution or university for the award of any degree, diploma, or any other academic recognition.

All the sources of information and reference materials used in the preparation of this project have been duly acknowledged.

Signature of the Candidate:

ACKNOWLEDGEMENT

It is with profound respect and deep gratitude that we acknowledge the guidance and support extended to us during this project.

We are sincerely thankful to **Amritsar Group of Colleges** for providing us with the opportunity to undertake this project, which is a part of the academic requirement for the degree of **Bachelor of Technology in Computer Science and Engineering**.

We wish to express our heartfelt appreciation to our project guide, **Er.Tejinder Sharma (Associate Professor)**, for his invaluable guidance, continuous encouragement, and unwavering support throughout the duration of this project. His insights, suggestions, and critical feedback have significantly contributed to the successful completion of this work.

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We extend our thanks to all the faculty members of the department for their assistance and cooperation. We also acknowledge the role of the lab staff and administrative personnel whose timely help and technical support enabled us to carry out this project without hindrance.

Finally, We would like to express our gratitude to our family and friends for their constant encouragement, patience, and understanding during this endeavour.

Kanwaldeep Singh (2233626)

TABLE OF CONTENTS

S. No.	Content	Page No.
1	Declaration	i
2	Acknowledgement	ii
3	List of figures	iii
4	List of collections	iv
5	Introduction	1
6	Requirement specification	3
7	Feasibility study	6
8	System analysis	8
9	System design	11
10	Testing & evaluation	16
11	Deployment & maintenance	19
12	Project outcomes	22
13	Conclusion and future work	33
14	Bibliography	35

LIST OF FIGURES

S. No.	Content	Page No.
1	DFD Level 0	10
2	DFD Level 1	10
3	Splash Screen	22
4	Dashboard Screen	22
5	Login Screen	23
6	Forgot Password Screen	23
7	Info Hub Screen	24
8	Blogs Screen	25
9	Profile Screen	25
10	My Team Screen	26
11	Team Orders Screen	26
12	Order Details Screen	27
13	My Customers Screen	27
14	Notifications Screen	28
15	CRM Dashboard Screen	28
16	Daily Plan Screen	29
17	My Visits Screen	29
18	My Follow Ups Screen	30
19	Top Planners Screen	30
20	Rishi Pukhraj Screen	31
21	Sale Cart Screen	32

LIST OF COLLECTIONS

S. No.	Content	Page No.
1	Admin Users Collection	12
2	Customers Collection	13
3	Orders Collection	13
4	App AI Chat Collection	14
5	Notifications Collection	14

INTRODUCTION

Background:

In today's rapidly growing digital healthcare and wellness industry, efficient management of customer relationships, product orders, and team coordination has become extremely important. Many healthcare and wellness organizations still rely on manual processes for maintaining customer records, handling orders, and managing communication between staff and customers. These traditional methods often lead to inefficiency, delayed updates, poor tracking of customer interactions, and lack of centralized management. As businesses expand, it becomes increasingly difficult to monitor team activities, manage CRM operations, and provide smooth customer service using conventional systems. Therefore, there is a strong need for a modern digital platform that can automate these processes and improve overall operational efficiency.

Advancements in mobile application development and backend technologies have created opportunities to build intelligent and user-friendly management systems. Modern applications can provide real-time updates, centralized data handling, secure authentication, and seamless communication between different user roles. Such systems help organizations improve productivity, reduce manual workload, and deliver better customer experiences through automation and smart workflow management.

The project **Smart Pukhraj** is designed to utilize these technological advancements for healthcare and wellness business management. It is a mobile application developed for Pukhraj Healthcare Pvt. Ltd. that enables Branch, Staff, and Customers to manage CRM activities, product orders, billing, and notifications through a centralized platform. The system includes features such as role-based login, customer management, order placement, QR and cash billing, CRM dashboards, and AI-based diet recommendations through Rishi Pukhraj.

Smart Pukhraj is developed using Flutter for creating a modern and interactive mobile application interface, PHP for backend API development, and MySQL (phpMyAdmin) for secure database management. GetX is used for state management and efficient navigation within the application. The integration of CRM functionalities, digital billing, and AI-powered wellness guidance enhances the overall capability of the system. The application aims to simplify healthcare business operations, improve coordination among users, and provide a reliable digital solution for customer and order management.

Objective:

The main objective of the **Smart Pukhraj** application is to develop an automated and user-friendly healthcare and wellness management platform that:

- Develop a centralized mobile application to manage CRM activities, customer records, and product orders efficiently.
- Implement role-based access for Branch, Staff, and Customer users with secure authentication.
- Provide real-time order management and billing system with QR and cash payment options.
- Enable efficient CRM tracking through daily plans, visits, follow-ups, and performance monitoring.
- Ensure secure storage and management of user and order data using PHP and MySQL database integration.
- Improve coordination, productivity, and customer experience through digital workflow management and AI-based wellness support.

Scope:

This project focuses on developing a comprehensive **Smart Pukhraj** healthcare and wellness management application with the following features:

- CRM Management: Branch and Staff users can manage customer interactions, daily plans, visits, and follow-ups efficiently through a centralized CRM system.
- Role-Based Access System: The application provides separate access and functionalities for Branch, Staff, and Customer users according to their roles and permissions.
- Order & Billing Management: Users can place product orders, manage cart details, and complete billing using QR or cash payment methods.
- Real-Time Notifications: The system provides updates and alerts related to orders, CRM activities, and customer operations for better coordination.
- AI-Based Wellness Guidance: The Rishi Pukhraj feature generates personalized diet charts and product suggestions based on customer health requirements.
- Future Scalability**:** The application can be expanded with advanced analytics, online payment gateways, multi-branch management in future versions.

REQUIREMENT SPECIFICATION

The Requirement Specification document outlines the essential features, functionalities, and characteristics required for the successful implementation of the **Smart Pukhraj** healthcare and wellness management application. This application is designed to streamline and automate CRM operations, customer management, product ordering, billing, and communication between Branch, Staff, and Customer users. Smart Pukhraj ensures a secure, efficient, and user-friendly experience for managing healthcare business operations digitally.

Functional Requirements

The functional requirements define the key features and operations that the Smart Pukhraj application must perform to meet its objectives. These requirements describe how the system behaves, what functionalities it provides to users, and how it interacts with databases and application modules.

User Authentication

- The system must allow Branch, Staff, and Customer users to log in securely using registered credentials such as mobile number, email, and password.
- The application should provide role-based authentication and access control according to user level IDs.
- Only authenticated users should be able to access protected features and personalized dashboards.

CRM Management Module

- The system should allow Branch and Staff users to manage customer interactions, visits, follow-ups, and daily planning activities.
- Branch users should be able to monitor staff performance, top planners, and team activities through CRM dashboards.
- Staff users should be able to add customer visit details, update follow-up status, and manage customer relationships efficiently.

Order & Billing Module

- The application should provide an efficient order and billing system that allows users to create product orders and complete transactions smoothly. The system should verify customer details using mobile numbers and support QR as well as cash payment methods for secure billing operations.

Dashboard and Visualization

- The system should generate organized dashboards and visual summaries of CRM activities, visits, and notifications. Using structured layouts and interactive sections, the dashboard should help Branch and Staff users monitor workflow, customer interactions, and team performance efficiently.

Recommendation System

- Based on customer wellness requirements, the system should provide personalized diet charts and healthcare product recommendations through the Rishi Pukhraj AI feature.
- The recommendation system aims to improve customer wellness awareness by helping users make better healthcare and lifestyle decisions through intelligent suggestions.

Non-Functional Requirements

The non-functional requirements define the quality attributes and performance standards that the Smart Pukhraj healthcare management application must meet to ensure efficient, secure, and user-friendly operation.

Security

- User data privacy and system security are of utmost importance. The application should implement secure authentication and role-based authorization for Branch, Staff, and Customer users.
- The application should also prevent common security threats such as SQL Injection, unauthorized access attempts, and Cross-Site Scripting (XSS).

Usability

- The user interface should be simple, interactive, and visually appealing to improve user engagement and healthcare workflow management. The dashboard should display CRM activities, orders, visits, and notifications in a clear and easy-to-understand format.
- The color scheme and glassmorphism design should create a modern and user-friendly experience. The application should be compatible with Android devices for better accessibility.

Performance

- The system should be optimized to provide quick responses to user actions such as logging in, customer verification, order placement, and dashboard loading. Average response time should remain minimal for standard operations.
- The backend APIs and database queries must handle multiple concurrent user requests efficiently without affecting performance. Data retrieval for customer records, CRM activities, and order history should remain fast and reliable.

Reliability and Availability

- The application must be reliable and capable of maintaining consistent functionality under normal operating conditions. The system should ensure proper synchronization of customer records, orders, and CRM activities even during network interruptions or unexpected failures.
- The system should remain available whenever required, allowing users to access the platform smoothly. Maintenance activities and updates should be managed properly to minimize downtime.

Scalability

- The system should be designed to handle increasing numbers of users, orders, and customer records without significant performance degradation. The backend architecture using PHP and MySQL should support future expansion and additional features.
- MySQL's flexible database structure – efficient management of customer records.

FEASIBILITY STUDY

A feasibility study is conducted to determine whether the proposed Smart Pukhraj healthcare and wellness management application is practical, achievable, and beneficial within the available resources, time, and technology. It helps identify the strengths, limitations, opportunities, and challenges associated with the development and deployment of the application. The feasibility study ensures that the system is technically, operationally, and economically suitable before full-scale implementation.

Technical Feasibility

The technical feasibility evaluates whether the available technologies and development tools are sufficient to develop and maintain the Smart Pukhraj application.

- The project uses modern and reliable technologies such as Flutter, PHP, MySQL, and GetX, which are widely supported and suitable for mobile application development.
- The system integrates CRM management, order handling, notifications, and AI-based wellness recommendations efficiently within a single application platform.
- The selected technologies provide fast performance, smooth UI handling, secure database management, and scalable application architecture.
- The overall system design ensures secure data handling, responsive dashboards, efficient API communication, and smooth integration between frontend and backend modules.

Thus, the project is technically feasible considering the available technologies, resources,and development expertise.

Operational Feasibility

Operational feasibility determines whether the application will function effectively after deployment and whether users will easily adapt to the system.

- The application provides a user-friendly and interactive interface that requires minimal training for Branch, Staff, and Customer users.
- Users can easily log in, manage CRM activities, create orders, monitor customer records, and complete billing operations through organized dashboards.

- The platform provides AI-based wellness recommendations, notifications, and efficient workflow management to improve healthcare business operations.
- The workflow—from login and CRM handling to order placement and billing—is smooth, efficient, and easy to manage for different user roles.
- The system can be operated effectively by users having basic smartphone and mobile application knowledge.

Since the application improves operational efficiency and simplifies healthcare workflow management, it is operationally feasible.

Economic Feasibility

Economic feasibility analyzes the cost-effectiveness of the Smart Pukhraj application compared to the benefits it provides.

- Development costs are minimized because the project uses widely available technologies such as Flutter, PHP, MySQL, and GetX, which reduce dependency on expensive software solutions.
- The system operates on existing mobile infrastructure, minimizing additional hardware
and operational expenses.
- The application reduces manual workload, improves order management efficiency, and saves time and resources, resulting in long-term operational benefits.
- Future maintenance and enhancement costs remain manageable because of the stable technologies and modular application architecture used in the system.

Thus, the overall cost of developing and maintaining the Smart Pukhraj application is justified by its operational advantages and long-term business benefits, making the project economically feasible.

SYSTEM ANALYSIS

Problem Definition

Manual CRM Management:

Healthcare and wellness businesses often manage customer records, follow-ups, and team activities manually, making CRM operations time-consuming, unorganized, and difficult to maintain accurately.

Inefficient Order Handling:

Traditional order management processes lack automation, causing delays in product ordering, billing, customer verification, and order tracking.

Lack of Centralized Management:

Customer records, staff activities, orders, and billing information are often scattered across multiple systems or maintained manually, making monitoring and coordination difficult.

Absence of Real-Time Updates:

Conventional systems do not provide instant notifications or real-time synchronization between Branch, Staff, and Customers, leading to communication delays and workflow inefficiencies.

No Intelligent Wellness Guidance:

Traditional healthcare management systems do not provide AI-based diet charts or personalized healthcare product recommendations for customers.

Data Disorganization:

Customer information, CRM reports, visits, follow-ups, and order details are difficult to manage efficiently without a centralized digital platform, increasing the risk of data duplication and record mismanagement.

Privacy and security concern:

Manual systems and unsecured applications may expose sensitive customer data, billing records, and CRM information to unauthorized access or misuse.

Need for Automation and Intelligence:

There is a need for a smart healthcare management solution that automates CRM operations, order handling, billing, notifications, and customer management with minimal manual effort.

Objective of New System:

The goal of Smart Pukhraj is to develop a modern healthcare and wellness management application that enables Branch, Staff, and Customer users to manage CRM activities, customer records, product orders, billing operations, notifications, and AI-based wellness recommendations through a centralized digital platform.

System Environment

Frontend Technology – Flutter:

Flutter is used for building a modern, responsive, and interactive mobile application interface. It provides fast performance, smooth navigation, reusable widgets, and an attractive glassmorphism-based healthcare UI experience across Android devices.

Backend Technology – PHP:

PHP is used for handling server-side operations such as user authentication, customer verification, CRM management, order processing, notifications, billing operations, and API communication between the frontend and database.

Database – MySQL (phpMyAdmin):

MySQL is used as the centralized database system for storing customer records, CRM activities, product orders, billing details, notifications, and user information securely and efficiently.

User Authentication and Security:

The system includes secure login mechanisms with role-based authentication for Branch, Staff, and Customer users. Sensitive customer records, billing information, and CRM data are protected through secure backend communication and controlled access permissions.

User Roles:

There are three primary user roles:

- **Branch:** Can manage staff members, monitor CRM activities, view reports, handle customer records, and supervise team orders.
- **Staff:** Can manage customer visits, follow-ups, CRM activities, and product orders within assigned permissions.
- **Customer:** Can view orders, access wellness recommendations, manage profile details, and interact with healthcare services.

Device Compatibility:

The application uses a responsive Flutter-based design to ensure functionality and optimized performance across Android smartphones and tablets.

Tools Used:

The project was developed using Flutter as frontend, Php as backened & MySQL(PhpMyAdmin).

DFD

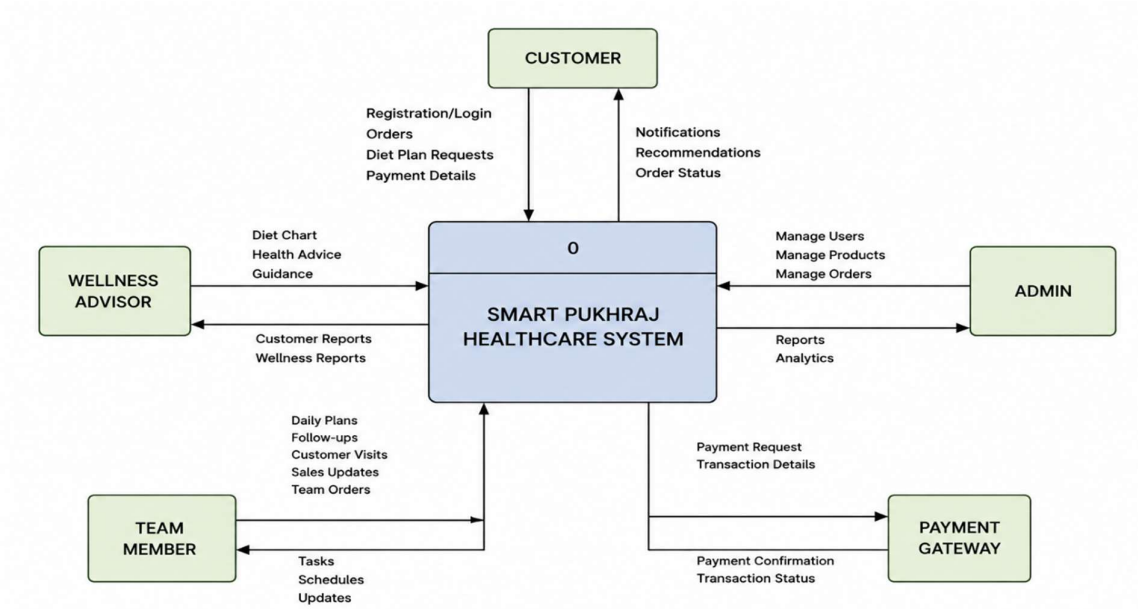


Fig 1 DFD Level 0

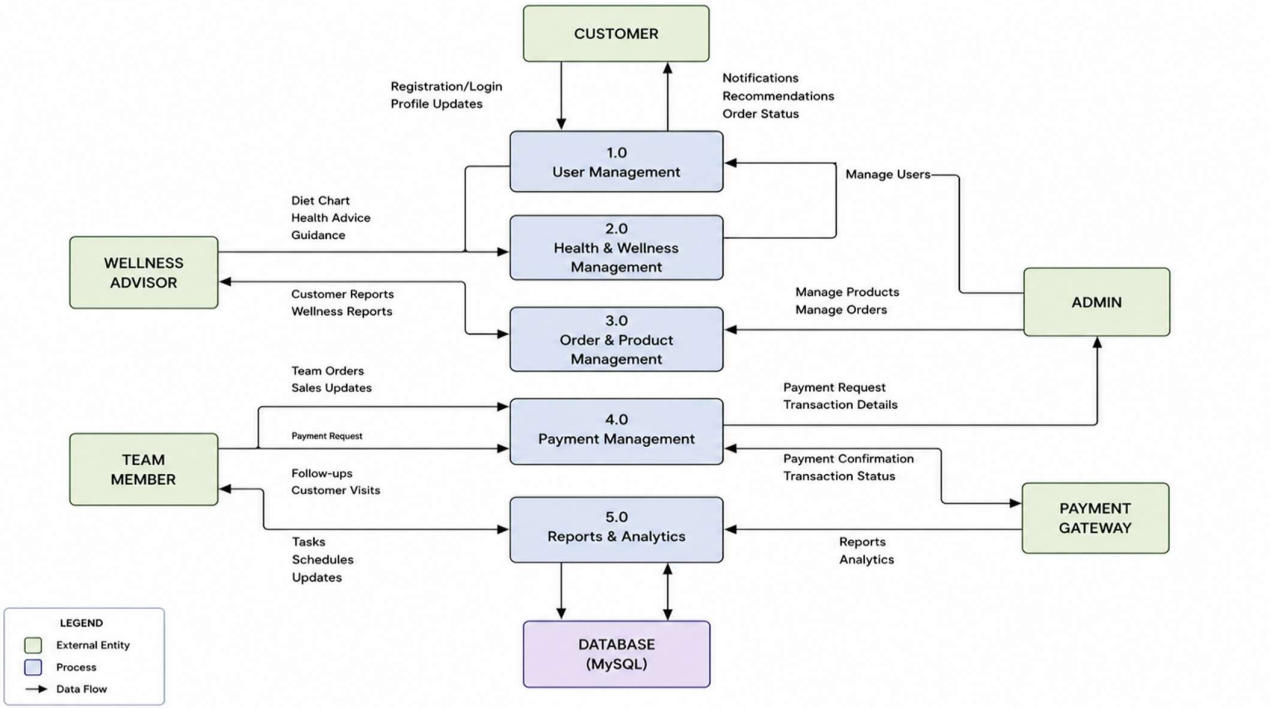


Fig No 2 DFD Level 1

SYSTEM DESIGN

Architecture

The Smart Pukhraj healthcare and wellness management application follows a modern three-tier client-server architecture that ensures efficient communication between users, the application server, and the database. The architecture is logically divided into three layers: Presentation Layer, Application Logic Layer, and Database Layer. This structure improves scalability, maintainability, security, and smooth workflow management across the system.

Presentation Layer (Frontend):

The frontend of Smart Pukhraj is developed using Flutter, which provides a responsive, modern, and interactive mobile application interface for smooth user interaction. The application follows a widget-based architecture with GetX state management to ensure fast navigation, organized UI handling, and efficient performance.

The frontend interface is designed with modern healthcare-themed glassmorphism UI elements, allowing Branch, Staff, and Customer users to access dashboards, CRM activities, notifications, orders, billing systems, and wellness features smoothly without performance issues.

Application Logic Layer (Backend):

The backend of the Smart Pukhraj system is developed using PHP, which provides a reliable and scalable server-side environment. This layer handles all business logic, processes API requests, and manages communication between the frontend and the database.

The backend manages functionalities such as user authentication, customer verification, CRM operations, order processing, notifications, billing management, and AI-based wellness recommendations. RESTful APIs are used for smooth data exchange between Flutter frontend screens and backend services while maintaining system performance and reliability..

Database Layer (Data Storage):

The database layer uses MySQL managed through phpMyAdmin for storing customer records, staff details, CRM reports, order history, billing data, notifications, and product information in a structured and organized manner.The relational database structure ensures efficient storage, fast data retrieval, secure management of records, and reliable handling of healthcare-related workflows.

Security Measures:

Security is an important aspect of the Smart Pukhraj application. The system uses role-based authentication for Branch, Staff, and Customer users, ensuring secure access control throughout the application. Sensitive customer information, billing records, and CRM data are protected using secure backend communication, controlled database access, and proper input validation techniques to prevent unauthorized access and data misuse.

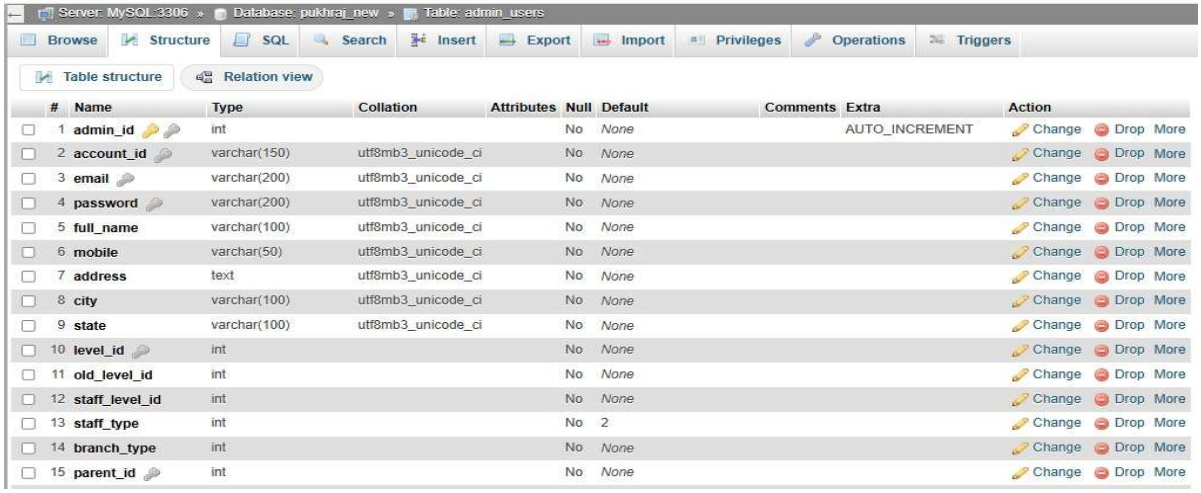
Database Design

The database plays a crucial role in managing customer records, CRM activities, product orders, billing operations and user management. MySQL database tables are designed efficiently to maintain structured healthcare data and support smooth application workflow.

Admin Users Collection:

This collection stores administrator and staff details for the Smart Pukhraj Healthcare System. Each record represents an authorized user system responsible for managing healthcare services. Fields include:

- **admin_id** – Unique identifier for each admin user.
- **email** – Registered email address of the admin user.
- **password** – Encrypted password used for secure authentication.
- **full_name** – Full name of the admin or staff member.
- **mobile** – Contact number of the user.
- **address** – Residential or office address of the user.
- **level_id** – Access level identifier defining user permissions.

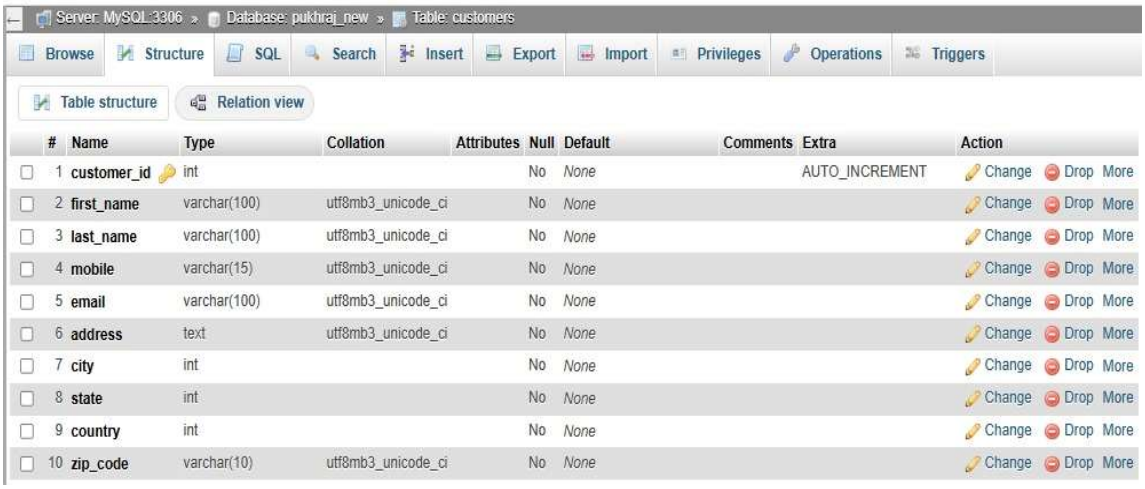


#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
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2	account_id	varchar(150)	utf8mb3_unicode_ci		No	None			Change Drop More
3	email	varchar(200)	utf8mb3_unicode_ci		No	None			Change Drop More
4	password	varchar(200)	utf8mb3_unicode_ci		No	None			Change Drop More
5	full_name	varchar(100)	utf8mb3_unicode_ci		No	None			Change Drop More
6	mobile	varchar(50)	utf8mb3_unicode_ci		No	None			Change Drop More
7	address	text	utf8mb3_unicode_ci		No	None			Change Drop More
8	city	varchar(100)	utf8mb3_unicode_ci		No	None			Change Drop More
9	state	varchar(100)	utf8mb3_unicode_ci		No	None			Change Drop More
10	level_id	int			No	None			Change Drop More
11	old_level_id	int			No	None			Change Drop More
12	staff_level_id	int			No	None			Change Drop More
13	staff_type	int			No	2			Change Drop More
14	branch_type	int			No	None			Change Drop More
15	parent_id	int			No	None			Change Drop More

Collection 1 – Admin Users Collection

Customers Collection:

This collection stores customer details and contact information for the Smart Pukhraj Healthcare System. Each record represents an individual customer associated with product orders, wellness guidance, and healthcare services.

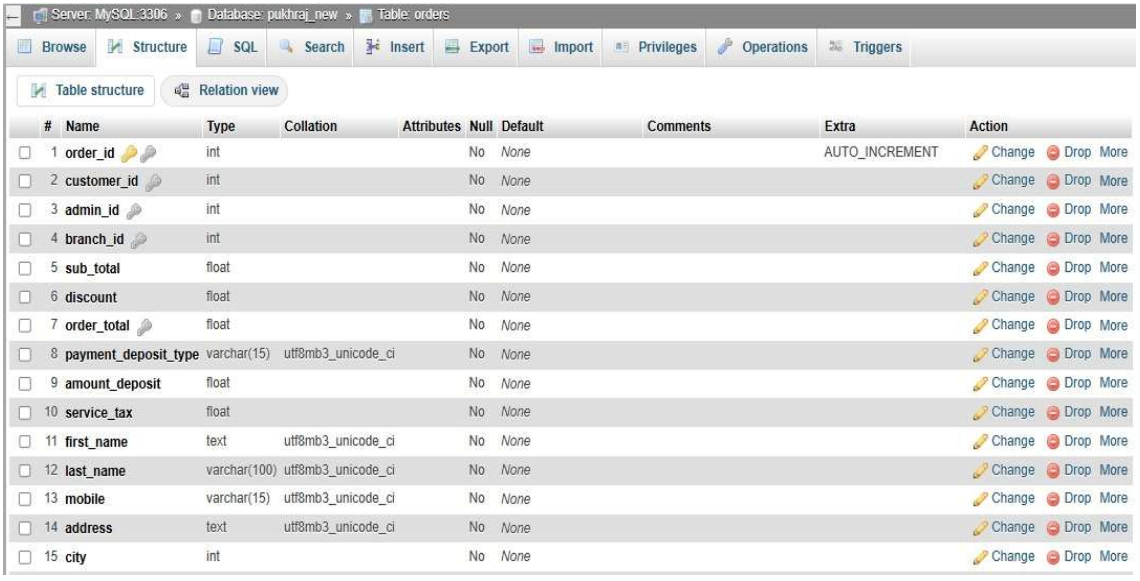


#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
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3	last_name	varchar(100)	utf8mb3_unicode_ci		No	None			Change Drop More
4	mobile	varchar(15)	utf8mb3_unicode_ci		No	None			Change Drop More
5	email	varchar(100)	utf8mb3_unicode_ci		No	None			Change Drop More
6	address	text	utf8mb3_unicode_ci		No	None			Change Drop More
7	city	int			No	None			Change Drop More
8	state	int			No	None			Change Drop More
9	country	int			No	None			Change Drop More
10	zip_code	varchar(10)	utf8mb3_unicode_ci		No	None			Change Drop More

Collection 2 – Customers Collection

Orders Collection:

This collection stores customer order and billing details for the Smart Pukhraj Healthcare System. Each record represents an order placed by a customer, including payment information, billing amounts, and delivery-related details.

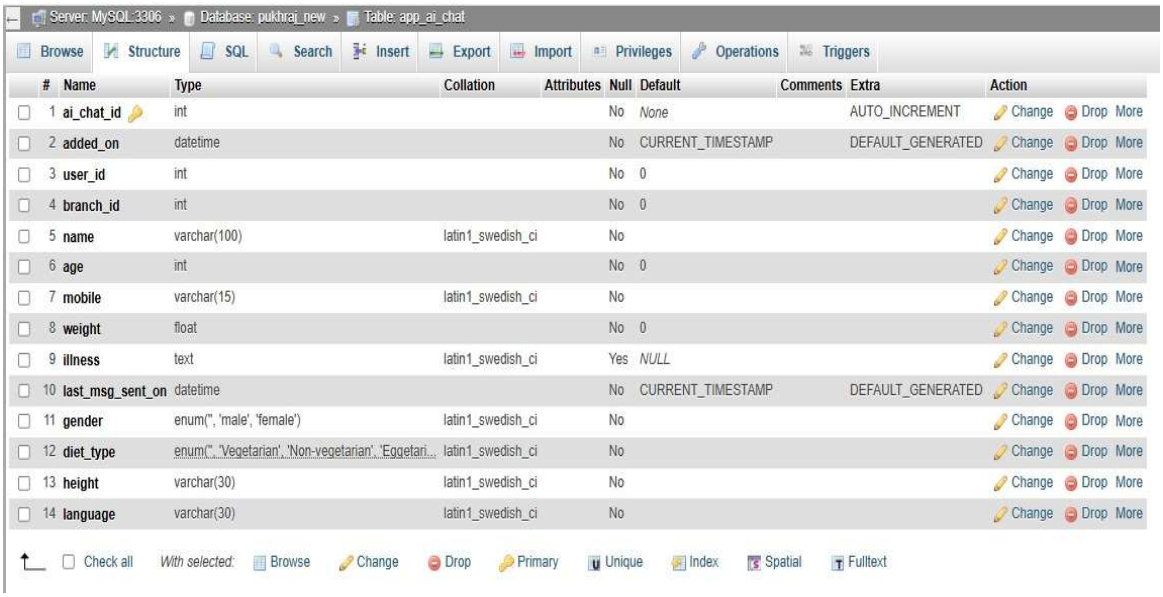


#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
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4	branch_id	int			No	None			Change Drop More
5	sub_total	float			No	None			Change Drop More
6	discount	float			No	None			Change Drop More
7	order_total	float			No	None			Change Drop More
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10	service_tax	float			No	None			Change Drop More
11	first_name	text	utf8mb3_unicode_ci		No	None			Change Drop More
12	last_name	varchar(100)	utf8mb3_unicode_ci		No	None			Change Drop More
13	mobile	varchar(15)	utf8mb3_unicode_ci		No	None			Change Drop More
14	address	text	utf8mb3_unicode_ci		No	None			Change Drop More
15	city	int			No	None			Change Drop More

Collection 3 – Orders Collection

App AI Chat Collection:

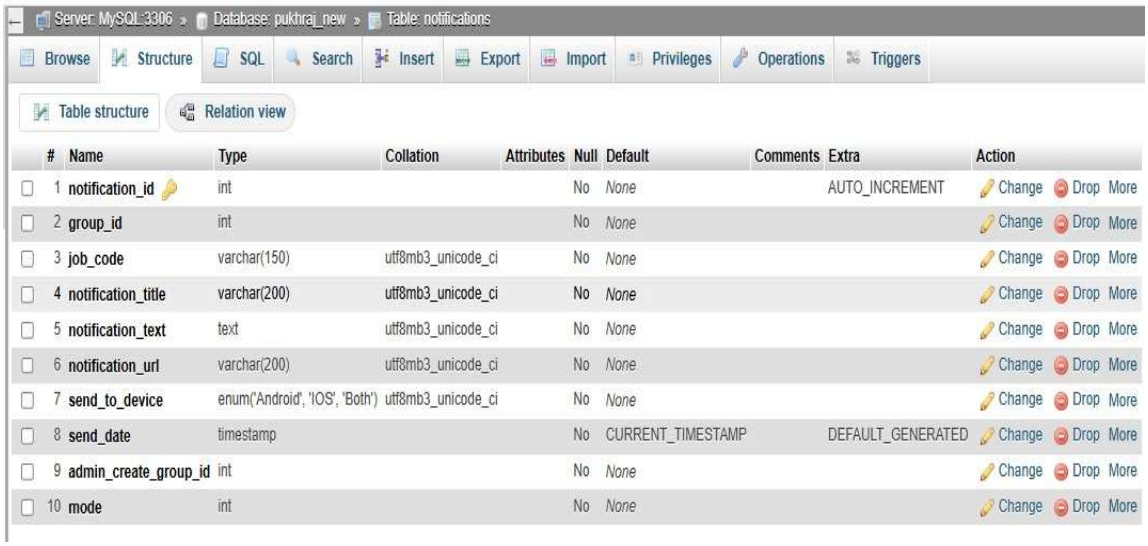
This collection stores customer details and contact information for the Smart Pukhraj Healthcare System. Each record represents an individual customer associated with product orders, wellness guidance, and healthcare services.



Collection 4 – App AI Chat Collection

Notifications Collection:

This collection stores notification and announcement details within the Smart Pukhraj Healthcare System. Each record represents a notification sent to users regarding healthcare updates, promotions, reminders, or system activities.



Collection 5 – Notifications Collection

Wireframes

Wireframes illustrate the structure and workflow of the Smart Pukhraj healthcare and wellness management application, ensuring a smooth and user-friendly experience across different user roles and devices.

Login Page:

A clean and secure interface allowing Branch, Staff, and Customer users to log in using their credentials. After successful authentication, users are redirected to their respective dashboards. The screen also includes forgot password functionality for secure account recovery.

Home/Dashboard:

The user is greeted with healthcare updates, notifications, and quick access links to major modules within the Smart Pukhraj application:

- My Team (view staff members and team performance details)
- Team Orders (manage customer orders, billing, and order history)
- CRM Dashboard (manage daily plans, visits, follow-ups, and planner reports)

CRM Dashboard:

The CRM section allows Branch and Staff users to manage daily plans, follow-ups, customer visits, and performance tracking. Different sections display planner reports, visit history, follow-up records, and staff activity summaries for efficient workflow management.

Wellness & AI Section:

The Rishi Pukhraj AI feature provides intelligent diet chart generation and healthcare product recommendations based on customer wellness requirements. Users can also access workshop videos, healthcare blogs, and wellness-related content.

Profile Settings:

Users can update profile information, view personal details, manage account settings, and access role-specific options through an organized profile section integrated with the dashboard modules.

TESTING & EVALUATION

Testing Strategies

A comprehensive testing strategy was adopted to ensure that the Smart Pukhraj healthcare and wellness management application functions smoothly, securely, and accurately across all modules.

Unit Testing

Each module, such as user authentication, CRM management, customer creation, order placement, billing operations, notifications, and AI-based wellness recommendations, was tested individually. Functions were verified using both valid and invalid inputs to ensure accurate output and proper error handling.

Integration Testing

Integration between the Flutter frontend, PHP backend APIs, and MySQL database was tested to ensure seamless data flow. Example: when a Staff or Branch user creates an order, customer details are verified from the database, products are added to the cart, billing is completed, and the order history is updated instantly within the system.

System Testing

End-to-end testing was conducted to validate complete workflows — from login and dashboard access to CRM handling, customer management, order placement, QR/cash billing, notifications, and AI wellness recommendations. All integrated modules were verified for smooth navigation, correct synchronization, and consistent UI behavior.

User Acceptance Testing (UAT)

Real users tested the application under practical healthcare workflow conditions. Branch users monitored team reports, Staff users managed customer visits and orders, and Customers accessed wellness features and order history. Feedback focused on usability, responsiveness, workflow efficiency, and overall user experience, leading to further UI and performance improvements.

Security Testing

Thorough checks were performed on authentication and data protection mechanisms. User access was secured according to Branch, Staff, and Customer roles, while backend APIs and database operations were protected from unauthorized access. Proper validation and secure communication practices were verified to ensure data privacy and system security.

Performance Testing

The application was tested under multiple simultaneous user activities. Backend APIs, dashboard loading, CRM reports, notifications, and order processing were monitored for smooth performance. Flutter frontend optimization and efficient MySQL query handling ensured responsive performance without major delays.

Bug Reports

During testing, several bugs were identified and resolved to improve reliability and application performance:

- **Customer details not auto-filling properly during order creation** → fixed DB verification logic.
- **Order summary calculation errors during billing** → corrected cart total calculation methods.
- **Notification delays on dashboard updates** → optimized API response handling.
- **UI alignment issues on different Android screen sizes** → improved responsive Flutter layouts.
- **Slow loading of CRM reports and team orders** → optimized MySQL queries and backend processing.

User Feedback

The application was demonstrated to Branch managers, Staff members, and Customers to collect feedback regarding usability, workflow efficiency, and healthcare management experience. Users tested modules such as CRM dashboard, order placement, notifications, AI wellness recommendations, and billing operations, providing valuable suggestions for further improvements.

Positive Feedback

- Clean, modern, and user-friendly healthcare interface with smooth navigation.
- Fast order creation and efficient customer verification workflow.
- CRM dashboard and follow-up management greatly improved staff productivity.
- AI-based diet chart and product recommendations enhanced customer engagement.
- Smooth billing system with QR and cash payment options improved order handling efficiency.

Suggestions for Improvement

Users provided several ideas for enhancing Smart Pukhraj in future versions:

- Add online payment gateway integration for digital transactions.
- Include multilingual language support for broader accessibility.
- Add detailed sales analytics and graphical business reports.
- Implement advanced notification scheduling and reminders.
- Introduce customer loyalty and reward management features.
- Improve AI recommendations using customer health history and wellness patterns.

Evaluation

The Smart Pukhraj application was evaluated for performance, usability, workflow efficiency, and reliability. All major modules — CRM management, customer handling, order processing, billing operations, notifications, and AI wellness recommendations — operated effectively during testing conditions.

The application maintained smooth performance under multiple user activities, securely handled customer data, and successfully achieved its objective of simplifying healthcare business management and wellness operations.

Overall, Smart Pukhraj successfully met its project goals and demonstrated strong potential as a modern healthcare and wellness management platform for real-world business environments.

DEPLOYMENT AND MAINTENANCE

Hosting

The Smart Pukhraj healthcare and wellness management system is designed as a mobile-based application that users can access through Android devices. Although large-scale deployment is not yet implemented, the planned deployment strategy involves using reliable hosting solutions to ensure accessibility, security, and smooth performance.

For the frontend, developed using Flutter, the application can be deployed as an Android APK or distributed through secure enterprise deployment methods. The Flutter framework ensures smooth performance, responsive UI handling, and compatibility across multiple Android devices.

The backend, developed using PHP, can be hosted on cloud-based hosting platforms or dedicated web servers supporting PHP and MySQL services. This setup ensures smooth API communication, customer verification, CRM management, billing operations, and order processing with minimal downtime. Secure HTTPS communication can be enabled to protect customer and billing data during transmission.

For the database, MySQL managed through phpMyAdmin will be used for centralized cloud-based data management. The database system supports secure storage of customer details, CRM activities, orders, billing records, notifications, and user information while maintaining scalability and reliability.

Once fully deployed, this hosting setup will allow Smart Pukhraj to remain accessible, maintainable, and scalable for future healthcare business growth while supporting smooth updates and consistent application performance.

Release Process

The release process of the Smart Pukhraj application follows a structured approach to ensure that updates, new features, and bug fixes are delivered efficiently and reliably. This approach helps developers implement new features and resolve issues without affecting the stable application environment.

After development, each module undergoes proper testing before deployment. Features such as CRM management, customer verification, order placement, billing operations, notifications, and AI wellness recommendations are verified carefully in the testing environment to identify performance issues or workflow errors before release.

Once testing is completed successfully, the updated application version is deployed for production use. The release process also includes performance verification, UI testing, API validation, and database consistency checks to ensure application reliability and smooth user experience.

The application follows a planned release cycle for adding enhancements, feature improvements, and security updates. Emergency fixes can also be deployed quickly whenever critical issues are detected. This structured release approach helps maintain application stability, security, and consistent healthcare workflow operations.

Overall, the release process ensures safe, smooth, and reliable deployment of Smart Pukhraj updates while supporting future scalability and system enhancements without affecting existing functionality.

Maintenance Strategy

The maintenance strategy for the Smart Pukhraj healthcare and wellness management application ensures that the system remains reliable, secure, and updated after deployment. Regular maintenance activities include monitoring application performance, fixing bugs, improving UI responsiveness, updating dependencies, and implementing security improvements to ensure uninterrupted operations.

Database maintenance is an important part of the strategy. This includes regular backups of the MySQL database, optimization of database queries, monitoring storage usage, and ensuring secure handling of customer, CRM, and billing data. These activities improve system stability and prevent data loss as the number of users and records increases.

Application code maintenance follows modular and organized development practices, making it easier to update existing features, add new functionalities, or improve workflow modules. Version control systems help track application changes efficiently and provide rollback support whenever required. Proper documentation of APIs, database structures, and workflows also simplifies future maintenance and development.

Security maintenance is another major focus area. User authentication systems, backend APIs, and database access permissions are regularly monitored to protect sensitive customer information and billing records from unauthorized access. User feedback and support activities are also maintained to identify issues quickly and implement improvements based on real-world healthcare workflow requirements.

By following this structured maintenance strategy, Smart Pukhraj can continue to provide a secure, scalable, and user-friendly healthcare management platform for Branch, Staff, and Customer users.

PROJECT OUTCOME



Fig 3 Splash Screen

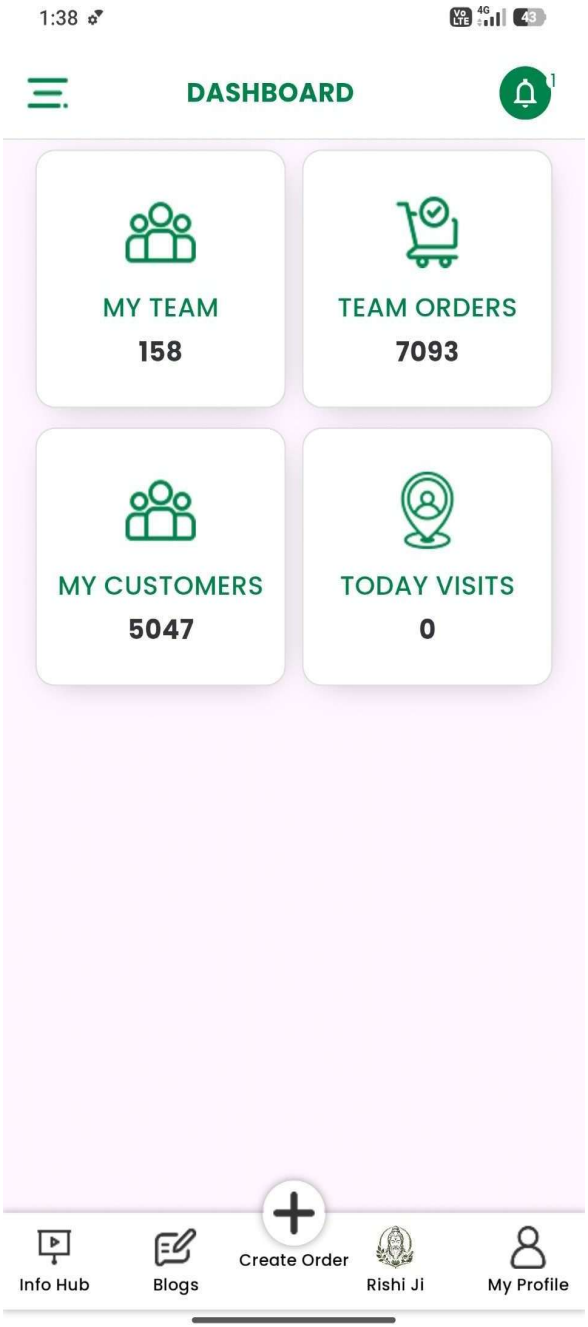


Fig 4 Dashboard Screen

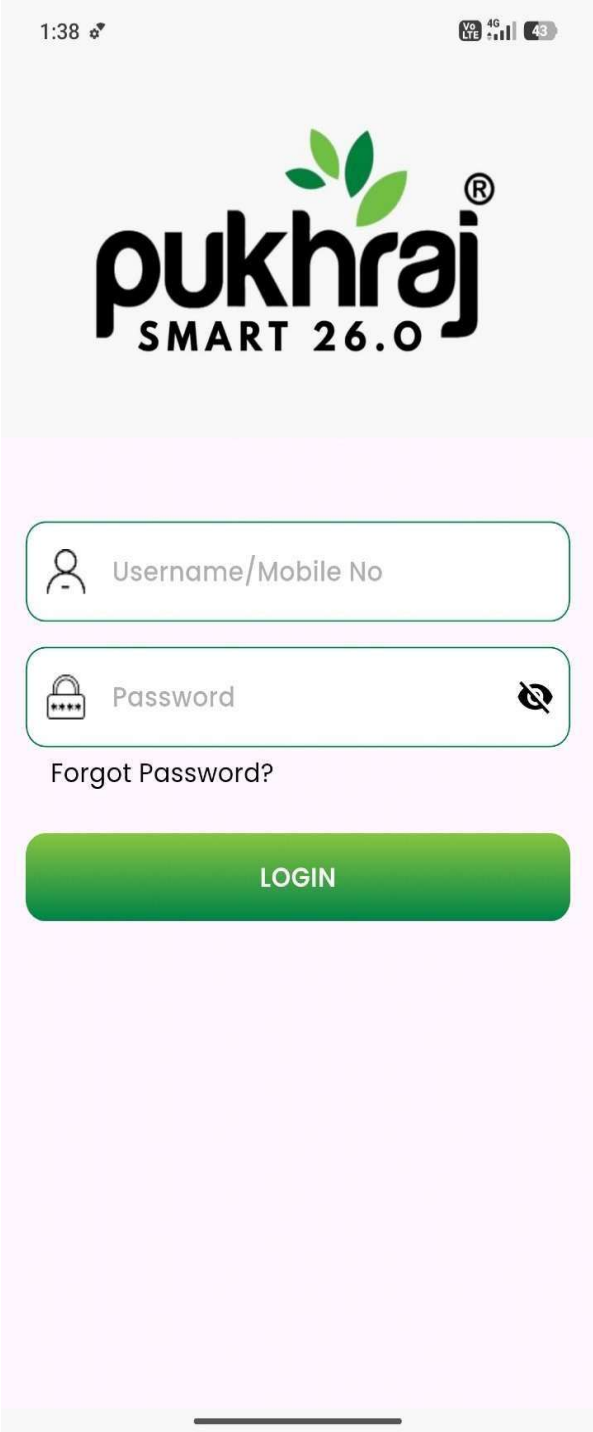


Fig 5 Login Screen

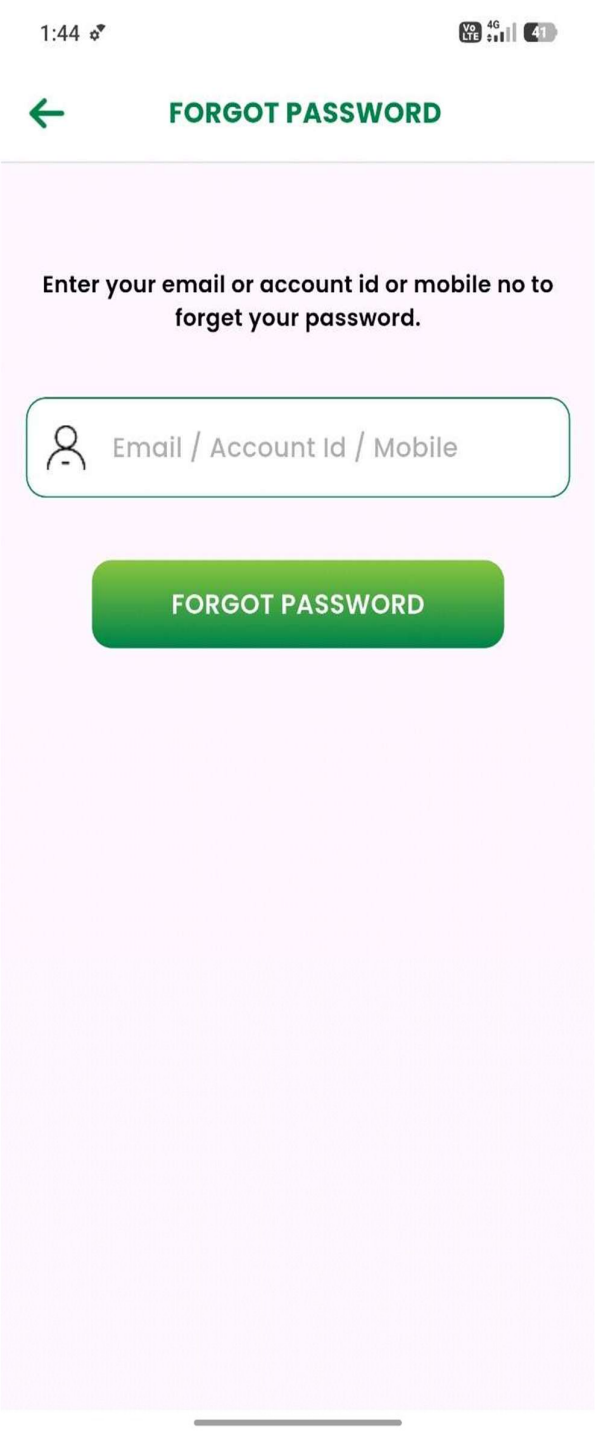


Fig 6 Forgot Password Screen

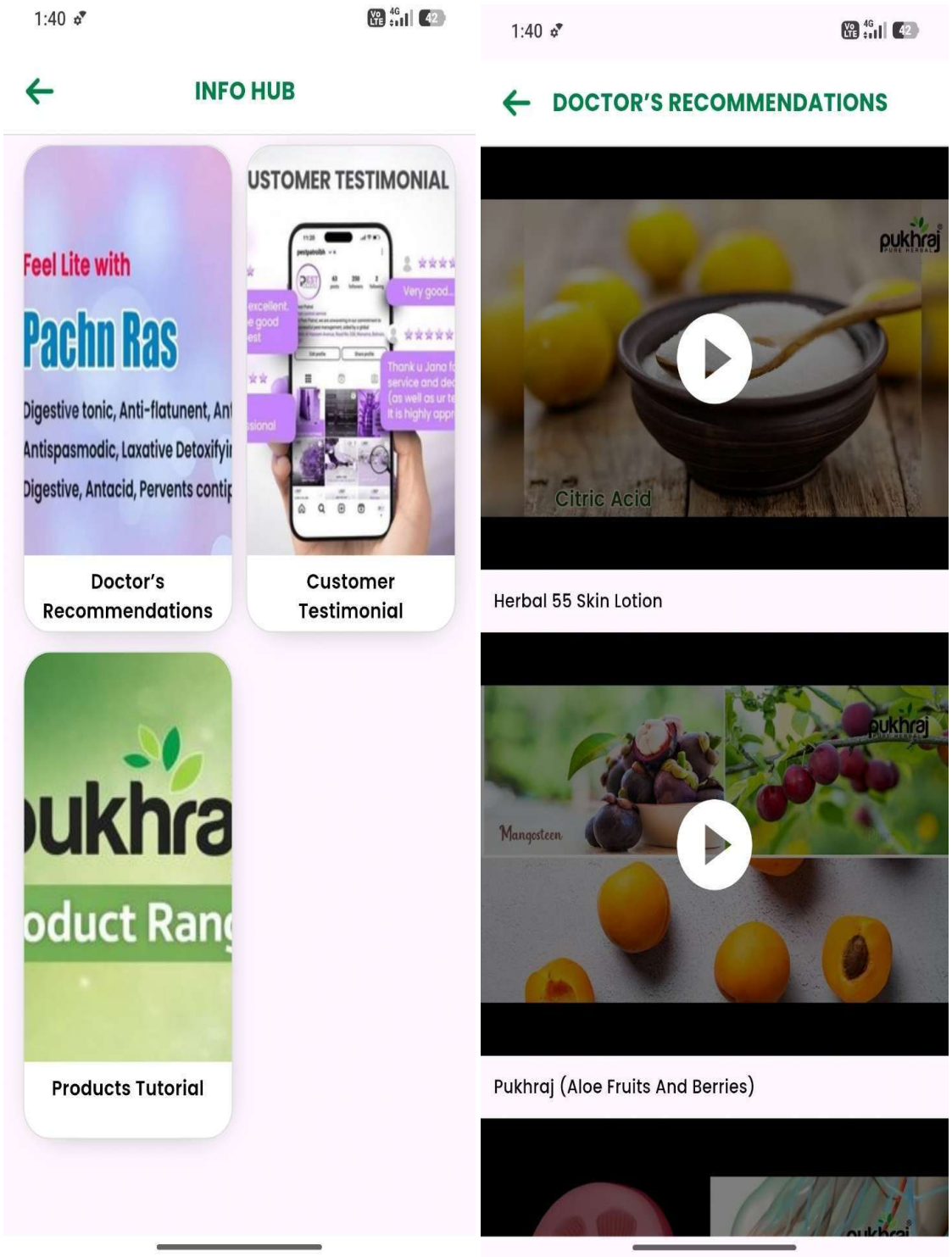


Fig 7 Info Hub Screen

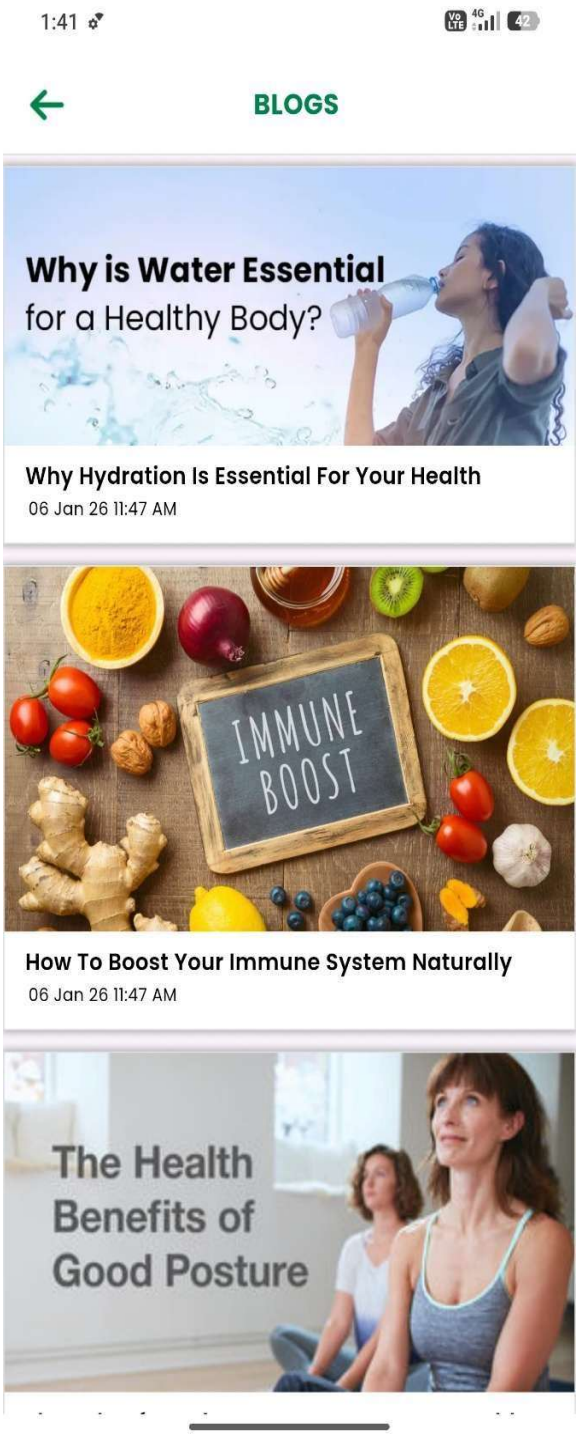


Fig 8 Blogs Screen

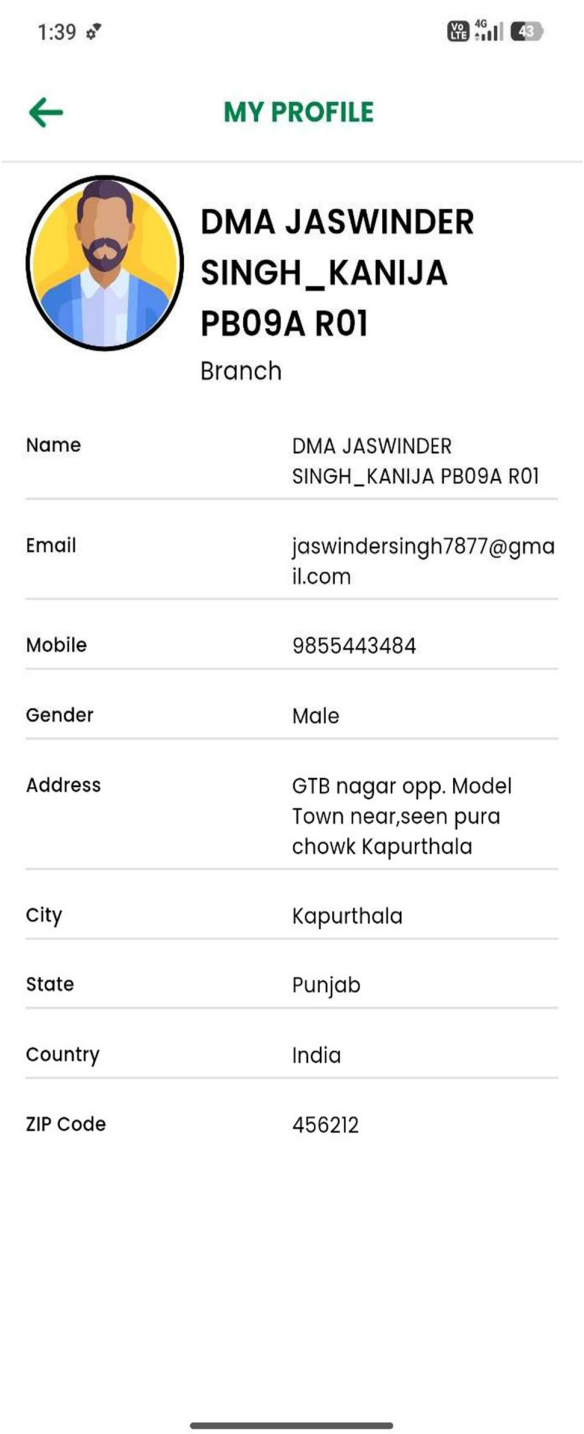


Fig 9 Profile Screen

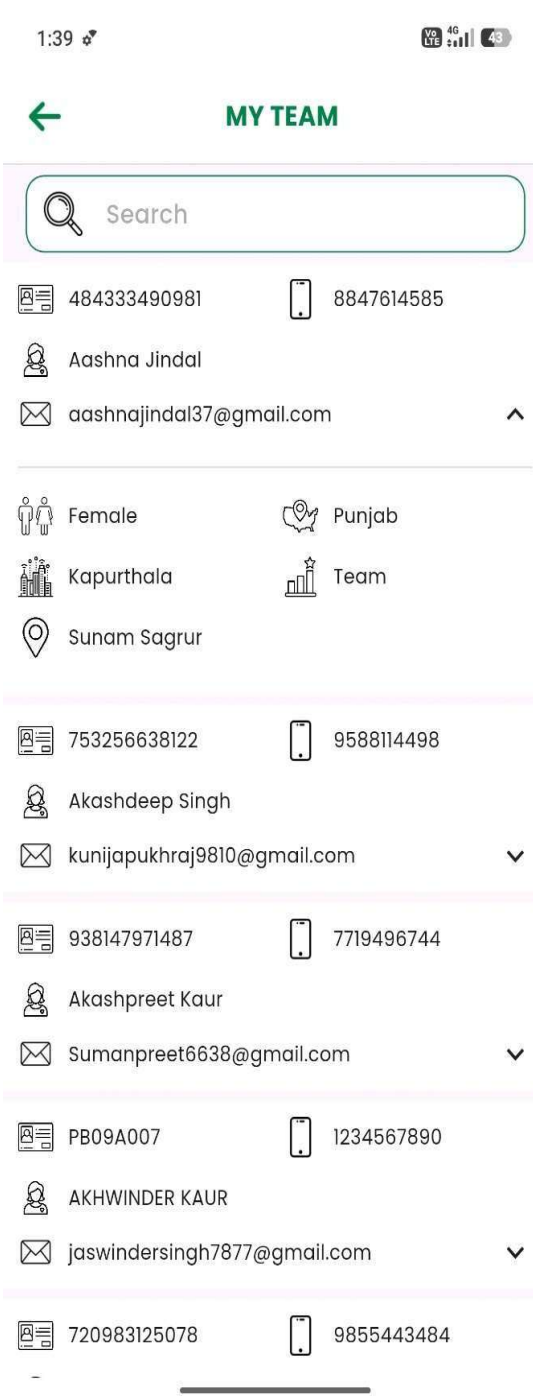


Fig 10 My Team Screen



Fig 11 Team Orders Screen

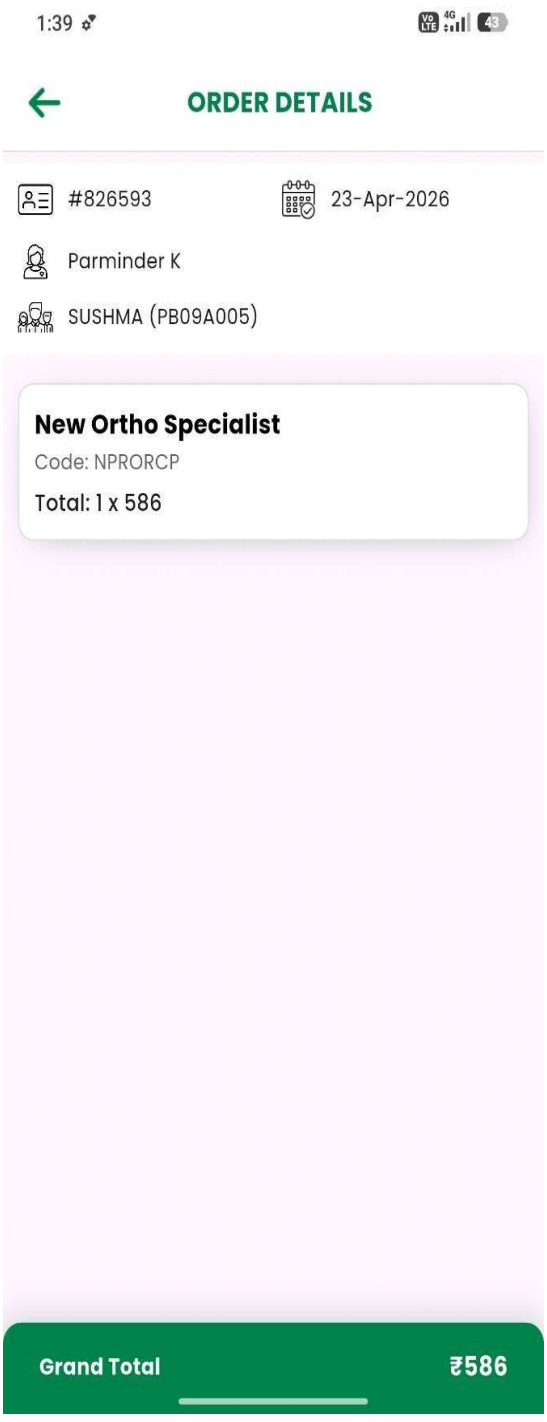


Fig 12 Order Details Screen



Fig 13 My Customers Screen

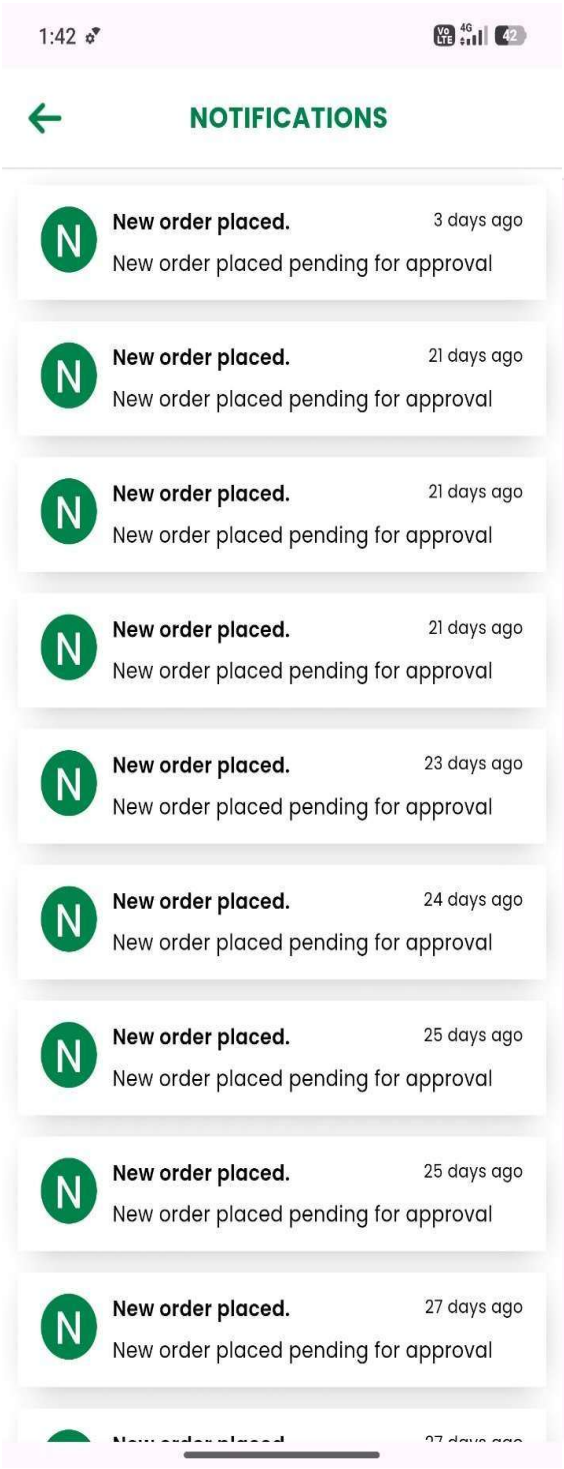


Fig 14 Notifications Screen

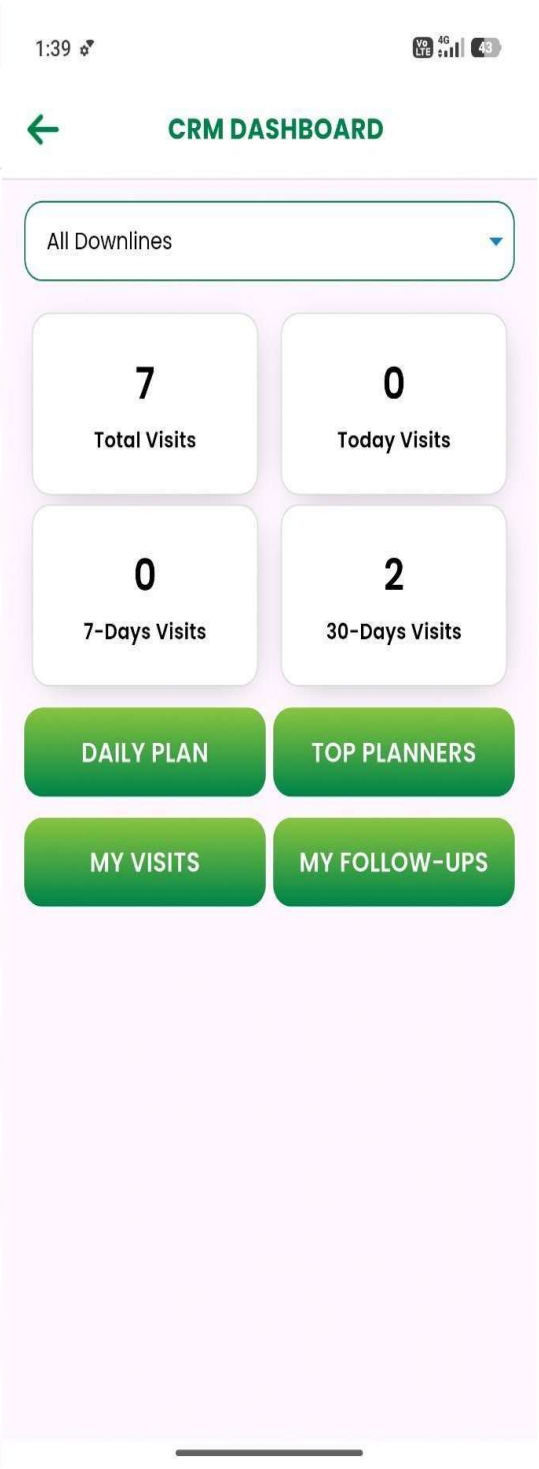


Fig 15 CRM Dashboard Screen

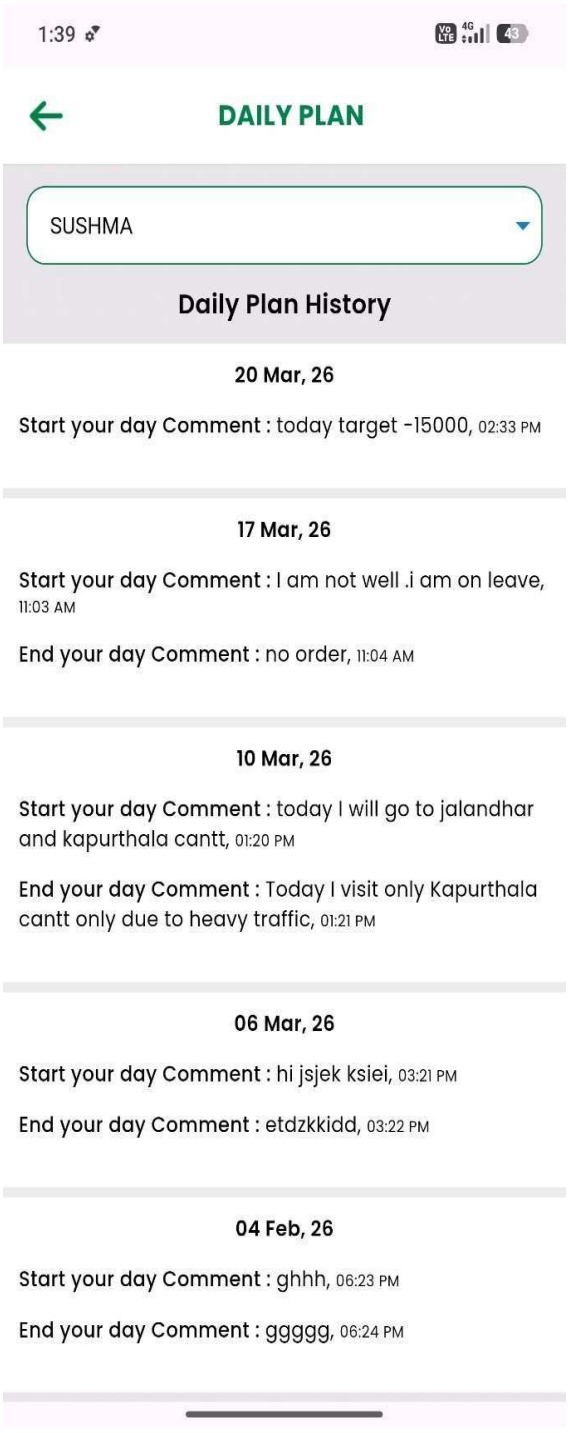


Fig 16 Daily Plan Screen

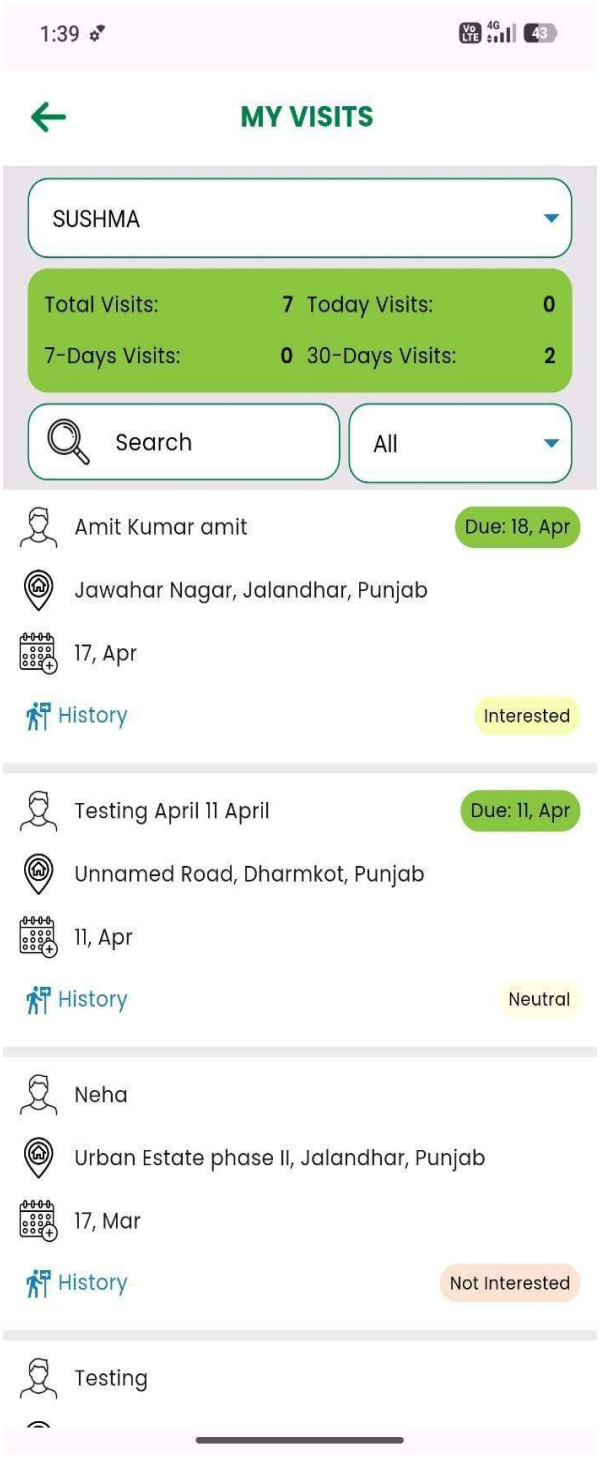


Fig 17 My Visits Screen

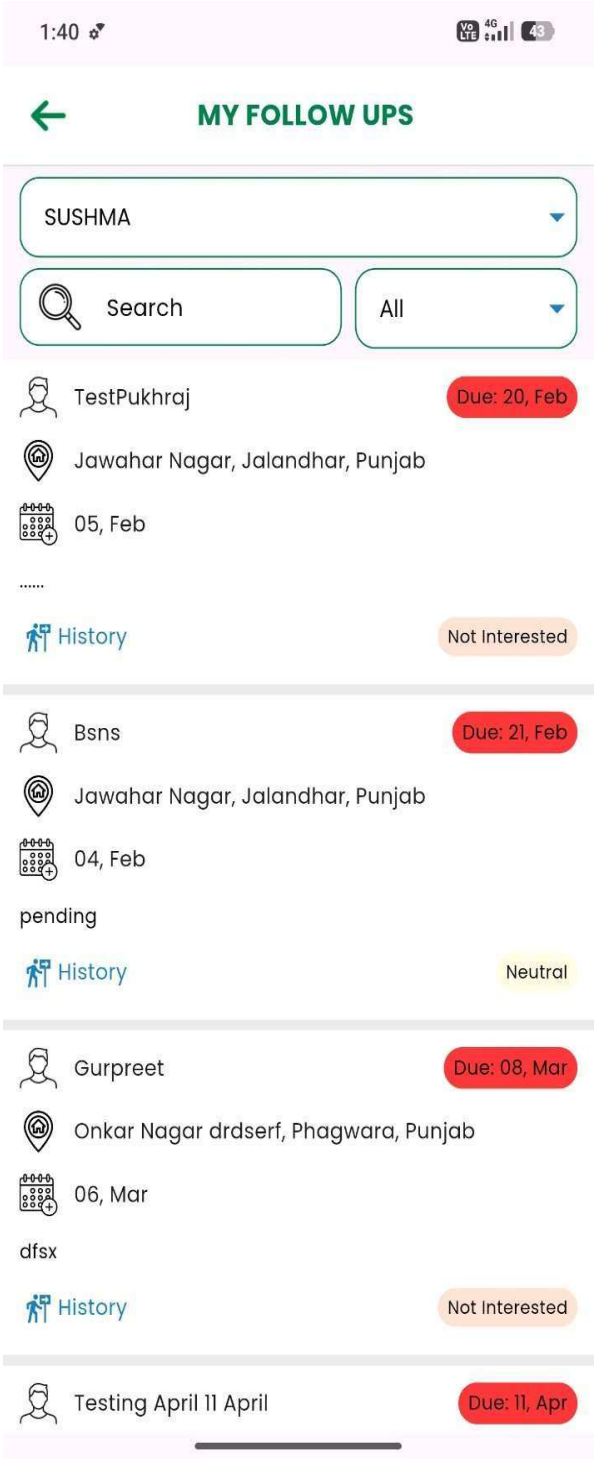


Fig 18 My Follow Ups Screen



Fig 19 Top Planners Screen

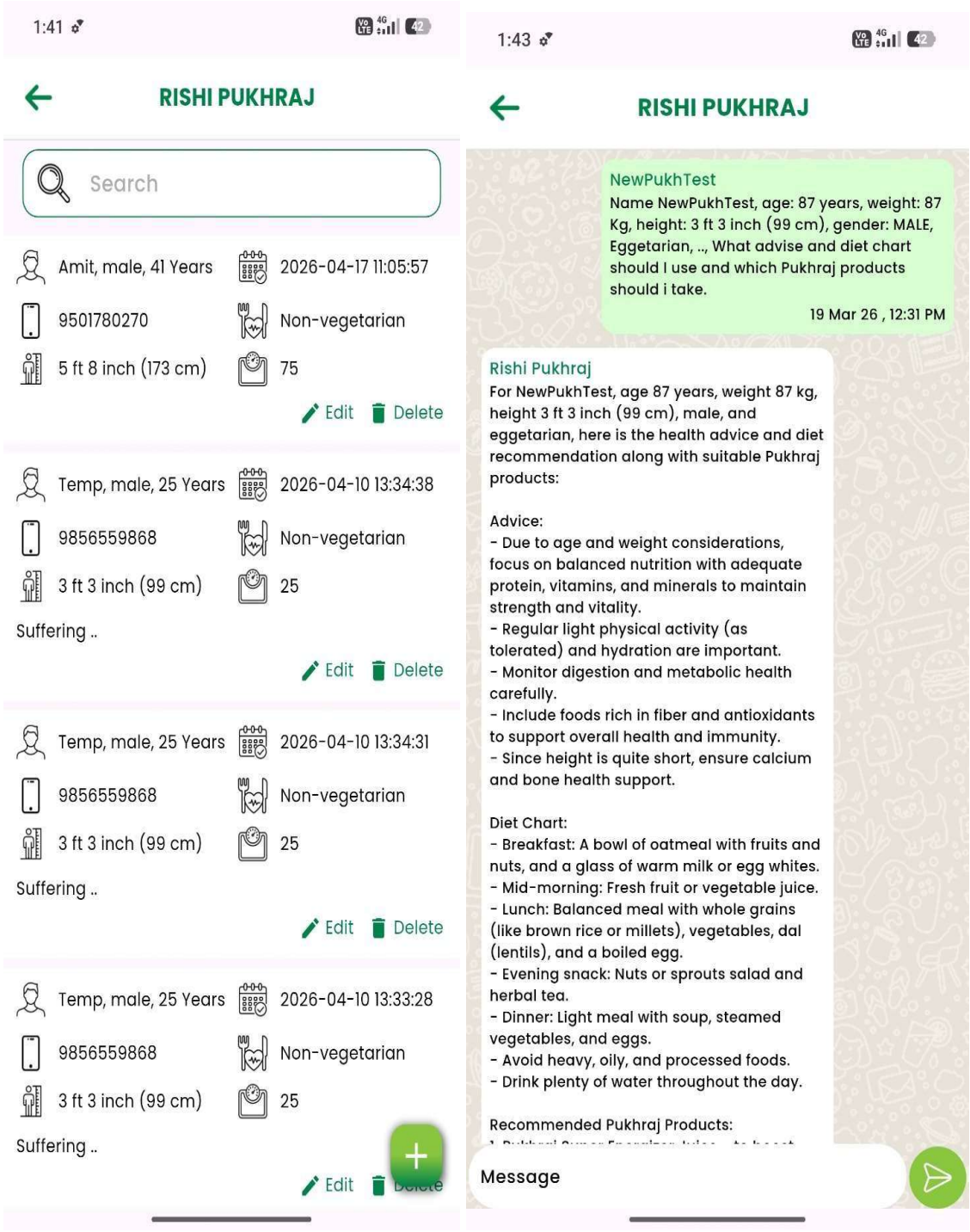


Fig 20 Rishi Pukhraj Screen

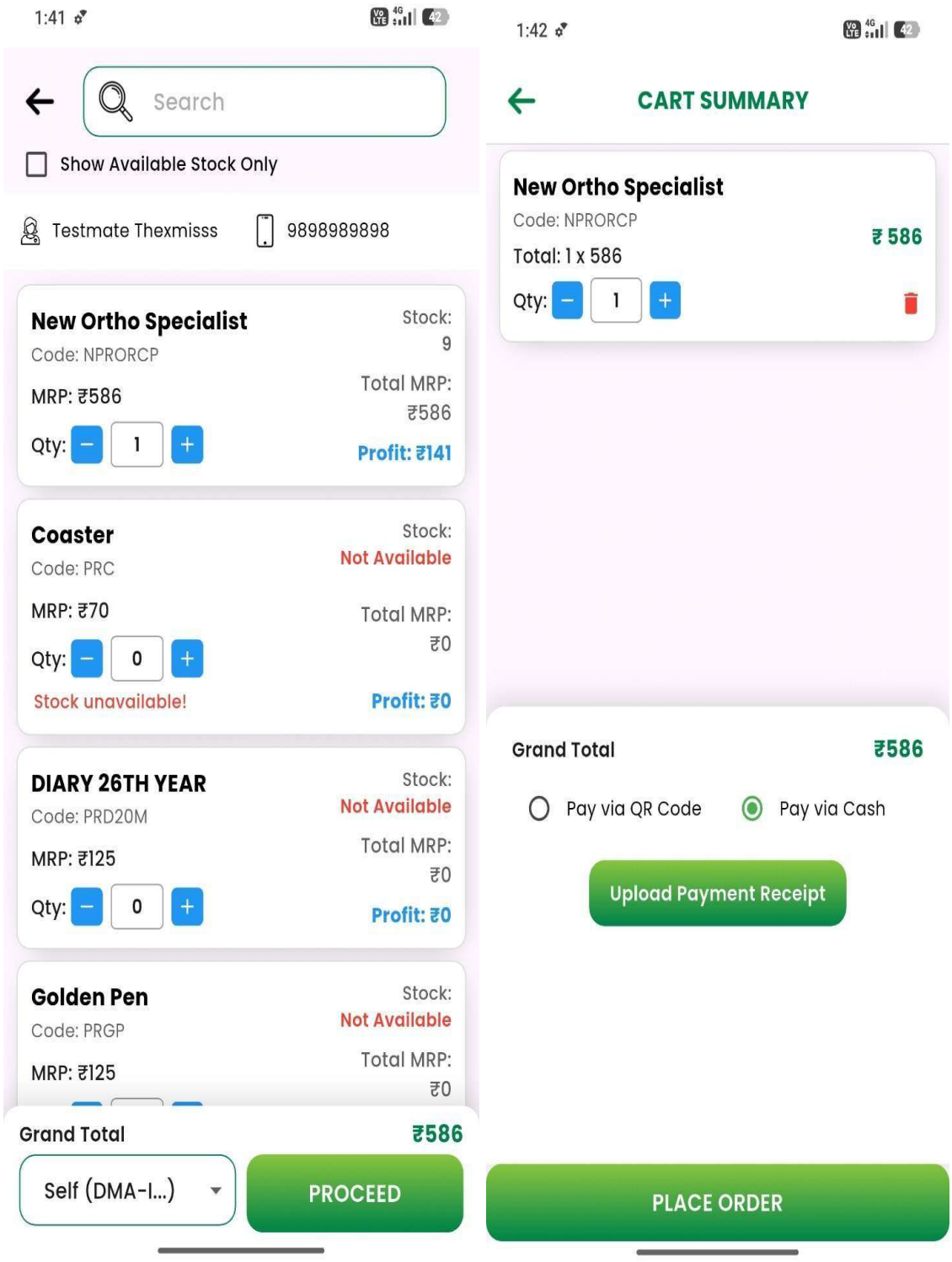


Fig 21 Sale Cart Screen

CONCLUSION AND FUTURE WORK

Achievements

The development of the Smart Pukhraj healthcare and wellness management application resulted in several important achievements in both technical and operational aspects. The project successfully delivered a fully functional mobile application that allows Branch, Staff, and Customer users to manage CRM activities, customer records, healthcare product orders, billing operations, notifications, and wellness-related services through a centralized digital platform.

The integration of Flutter for frontend development and PHP with MySQL for backend management enabled smooth system performance, secure data handling, and efficient communication between application modules. The application provides role-based authentication and organized dashboards, ensuring secure access and proper workflow management for different user roles.

A major achievement of the project is the implementation of CRM management, order creation, customer verification, and billing workflows within a single application environment. The system supports real-time notifications, customer record handling, team monitoring, and AI-based wellness recommendations through the Rishi Pukhraj feature, improving overall healthcare business efficiency.

The application also successfully integrates modern UI design using Flutter with GetX state management, resulting in smooth navigation, responsive screens, and an interactive healthcare-themed user experience. Optimized backend APIs, structured database management, and modular development practices contributed to application reliability, scalability, and performance.

Beyond technical implementation, Smart Pukhraj achieved the broader objective of simplifying healthcare business operations, improving customer management, and enhancing wellness service delivery through digital automation and intelligent healthcare solutions.

Scope for Future Enhancement

The Smart Pukhraj healthcare and wellness management application has strong potential for future improvements and feature expansion. One major enhancement area is the integration of advanced online payment gateways to support secure digital transactions directly within the application.

In the future, the AI-based Rishi Pukhraj recommendation system can be enhanced using advanced machine learning algorithms to provide more accurate diet charts, wellness guidance, and personalized healthcare product recommendations based on customer history and wellness patterns.

The application can also be extended with video consultation and telehealth support, allowing customers to communicate directly with wellness experts or healthcare professionals through online sessions. Integration of appointment booking and calendar scheduling modules can further improve healthcare workflow management.

Another important enhancement is the addition of multilingual support, making the platform accessible to a wider audience across different regions. Advanced analytics dashboards can also be implemented to provide Branch users with detailed business reports, customer insights, sales performance, and CRM statistics.

Future versions may also include customer loyalty programs, inventory management systems, barcode scanning, advanced notification scheduling, and cloud-based synchronization for improved scalability and operational efficiency.

Overall, the future scope of Smart Pukhraj lies in transforming it into a complete digital healthcare and wellness ecosystem that combines smart automation, AI-powered recommendations, customer engagement, and modern healthcare management solutions within a unified platform.

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